AGRIVIVE / DOBHA-CELL

JHARKHAND / DOBHA-INSPIRED / POND RUNOFF INTERCEPTOR

SERIES 10 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



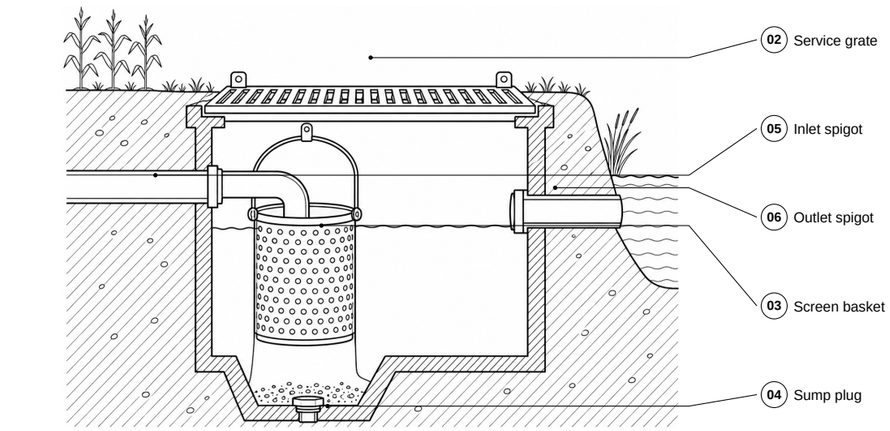
STATE CONTEXT / INDIGENOUS INSPIRATION

Dobhas are small farm ponds promoted in Jharkhand for local runoff collection. DOBHA-CELL is a new serviceable sediment interceptor ahead of a pond; it does not replace the pond or establish water quality.

Practice source: Jal Shakti groundwater harvesting best practices | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 160 L x 120 W x 140 H

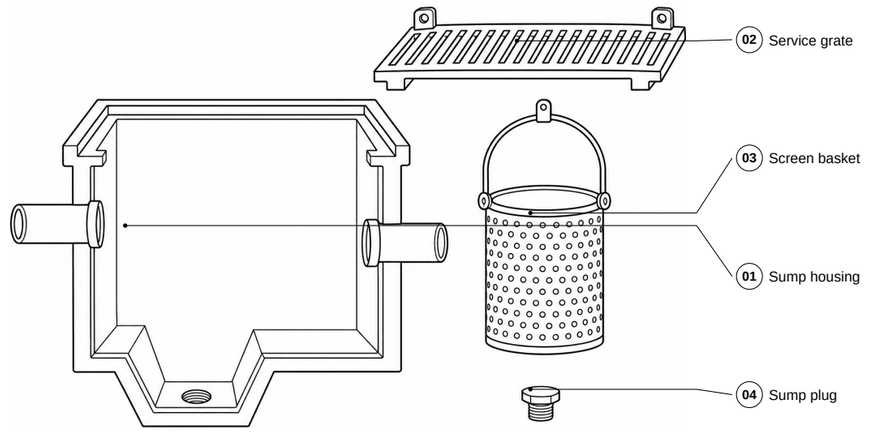


DESIGN / OPERATION NOTES

Set grate near grade and provide accessible service clearance.
Runoff enters the basket; water passes its perforations into the surrounding chamber.
Sediment can collect in the sump; remove basket and plug for cleaning.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
160 L x 120 W x 140 H

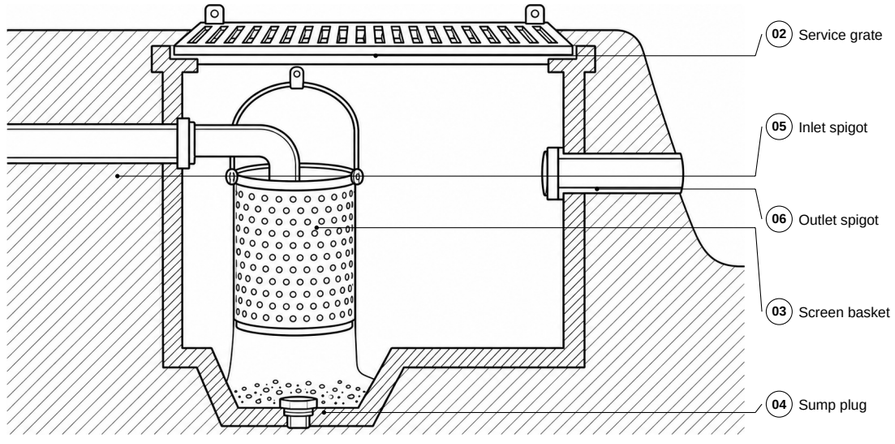


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

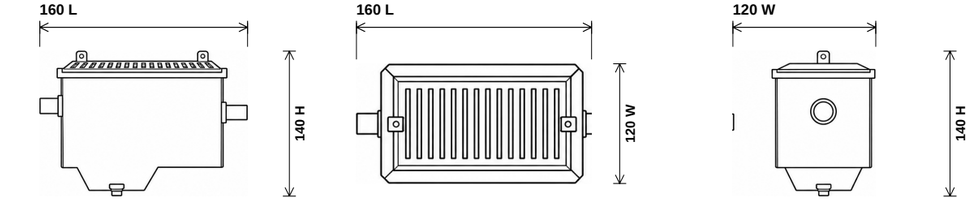
SECTION BASIS / mm
140 L x 100 W x 100 H clear envelope before basket



DESIGN / OPERATION NOTES

Runoff enters the basket; water passes its perforations into the surrounding chamber.
Sediment can collect in the sump; remove basket and plug for cleaning.

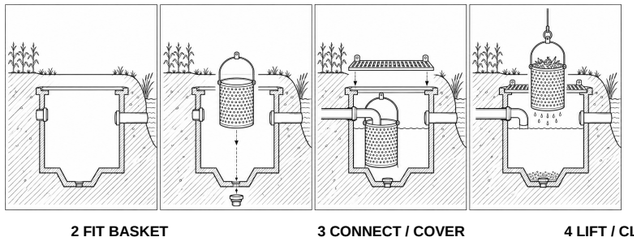
03 ORTHOGRAPHIC STUDIES / mm



FRONT **TOP** **SIDE**

External 160 L x 120 W x 140 H. Clear envelope 140 L x 100 W x 100 H before basket / freeboard.

05 INSTALLATION / OPERATING SEQUENCE



1 SEAT HOUSING **2 FIT BASKET** **3 CONNECT / COVER** **4 LIFT / CLEAN**

Prototype fit, access and operation require trials.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

External L x W x H	**160 x 120 x 140**
Clear envelope	**140 x 100 x 100**
Ports / basket detail	**TBD**

02 GEOMETRIC CHECK

GROSS CHAMBER ENVELOPE

1,400.0 mL

140 x 100 x 100 / 1000
Operating sediment storage/flow volume must be measured

03 PARTS / QTY

01 Sump housing	**1**
02 Service grate	**1**
03 Screen basket	**1**
04 Sump plug	**1**
05 Inlet spigot	**Integral**
06 Outlet spigot	**Integral**

Housing/grate/basket: candidate polymer. Plug seal and screen opening TBD.

04 INTERFACES / CHECKS

Check inlet/outlet inverts, freeboard and bypass.

Test screen clogging and basket withdrawal clearance.

Measure working sump volume with internals installed.

PRELIMINARY / VALIDATION PENDING

DESIGN DEVELOPMENT RECORD

01 / GEOMETRY
Nominal body sizes are proposed. Generated views are illustrative; small features and interfaces may vary.

02 / BEFORE FABRICATION
Develop one CAD model, assign wall thicknesses, tolerances and mating fits. Check print orientation and service access.

03 / TRIAL RECORD
Measure dimensions and retained volume or opening area. Record leakage, clogging and soil anchorage under observed conditions.

No discharge, treatment or durability performance is established.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-10 / P01	DOBHA-CELL	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	10 / 28